

Your grade: 96%

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Next item →

1. What is serialization in the context of machine learning?

1 / 1 point

- ☐ A way to visualize the performance of a machine learning model.
 - ☒ A method to convert a model into a format that can be easily saved and loaded.
 - ☐ A technique to improve the speed of training a machine learning model.
 - ☐ A process to increase the accuracy of a machine learning model.
- ☒ **Correct**
Correct! Serialization allows models to be saved and reused, which is essential for handling large datasets and deploying models.

2. How are digital images represented in the context of computer memory?

1 / 1 point

- ☐ As a series of text descriptions
 - ☒ As a set of numbers corresponding to pixel values
 - ☐ As a continuous function corresponding to light intensity
 - ☐ As a list of color names
- ☒ **Correct**
Correct! Digital images are represented as a set of numbers where each number corresponds to a pixel value.

3. Which of the following are valid steps in changing the color channel order of an image using Keras?

1 / 1 point

- ☒ Load the image as a PIL image
- ☒ **Correct**
Correct! Loading the image as a PIL image is the first step.
- ☒ Convert the image to a NumPy array using `image_to_array`
- ☒ **Correct**
Correct! Converting the image to a NumPy array is necessary to manipulate the color channels.
- ☒ Use the `array_to_image` function to convert the array back to a PIL image
- ☒ **Correct**
Correct! Converting the array back to a PIL image is the final step before saving.
- ☒ Change the order of the color channels in the NumPy array
- ☒ **Correct**
Correct! Changing the order of the color channels in the NumPy array is the key step.
- ☐ Save the NumPy array directly as an image file

4. What is the purpose of the `array_to_image` function in Keras?

1 / 1 point

- ☐ Enhance image contrast
 - ☐ Convert PIL images to grayscale
 - ☒ Convert NumPy arrays back to PIL-based images
 - ☐ Resize images
- ☒ **Correct**
Correct! The `array_to_image` function is used to convert NumPy arrays back to PIL-based images.

5. What are the key concepts involved in the representation of color images in array format?

0.8 / 1 point

- ☒ Color channels
- ☒ **Correct**
Correct! Color channels are fundamental in the representation of color images.
- ☒ Pixel intensity values
- ☒ **Correct**
Correct! Pixel intensity values are crucial in representing color images.
- ☒ Array dimensions
- ☒ **Correct**
Correct! Array dimensions are vital in representing the structure of color images.
- ☐ Image format (e.g., JPEG, PNG)
- ☒ Image resolution
- ☒ **This should not be selected**
Incorrect. While important, resolution is not a key concept in the array representation of color images.